Melika Baghi

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Research Interests

High-dimensional statistical learning; uncertainty quantification and conformal prediction; multimodal learning and missing data; sequential decision making and bandits; reduced-order and surrogate modeling; tensor and geometry-aware learning; scientific machine learning.

Education

Georgia Institute of Technology *Atlanta, GA*
Ph.D. in Industrial & Systems Engineering — System Informatics & Control *Aug 2023 – Dec 2027 (expected)*

- Advisors: Prof. Kamran Paynabar and Prof. Xiao Liu
- Dissertation: *Physics-Based Machine Learning for Fast Prediction of Engineering Simulations*

Georgia Institute of Technology *Atlanta, GA*
M.S. in Statistics *Aug 2023 – May 2026*

Amirkabir University of Technology (Tehran Polytechnic) *Tehran, Iran*
M.S. in Industrial & Systems Engineering — Healthcare Systems; GPA: 4.0 / 4.0 (19.35 / 20) *Oct 2021 – Jul 2023*

- Thesis: *Scheduling of Medical Tourists Visiting Medical Centers Using an Agent-Based Approach*
- Advisor: Prof. Hadi Mosadegh
- Direct-entry admitted as an elite student (ranked 3rd of 120 applicants)

Amirkabir University of Technology (Tehran Polytechnic) *Tehran, Iran*
B.S. in Industrial Engineering; GPA: 3.91 / 4.0 (18.65 / 20) *Oct 2018 – Jun 2021*

Amirkabir University of Technology (Tehran Polytechnic) *Tehran, Iran*
B.S. in Aerospace Engineering; GPA: 3.79 / 4.0 (18.14 / 20) *Sep 2016 – May 2020*

- Advisor: Prof. Mehdi Sabzeh Parvar
- Selected as an elite student and approved to pursue a dual degree in Aerospace Engineering (2018)

Publications and Manuscripts

Preprints and Manuscripts Under Review

[1] **M. Baghi**, X. Liu, K. Paynabar. "Constrained Extreme Gradient Boosting for Parametric Proper Orthogonal Decomposition Subspace Prediction." *Manuscript under review.* [arXiv:2605.04130] [code]

- Proposes cXGBoost, a constrained gradient-boosting method that predicts the proper orthogonal decomposition (POD) basis at new parameter settings: subspaces are mapped from the Grassmann manifold to a Euclidean space, and the norm of the predicted vectors is bounded so that the mapping stays injective. On four parametric problems (cylinder wake, Kolmogorov flow, the Kuramoto–Sivashinsky equation and a forced beam), cXGBoost has lower mean reconstruction errors than the projected Gaussian process and Grassmann interpolation.

[2] **M. Baghi**, H. Mosadegh. "Managing Shared Hospital Capacity for Medical Tourism: Patient Behavior and Hybrid Simulation." *Under review, Health Care Management Science.* [arXiv:2605.19139] [code] [DOI]

- Develops a hybrid ABS+DES model of a multi-specialty international patient department in AnyLogic that represents channel choice, physician switching, treatment abandonment, and medication-related deterioration. A 256-run two-level fractional factorial design screens 16 controllable factors across six performance measures and identifies bed capacity as the most consistent driver of waiting; priority rules, channel design, and appointment timing also matter. In a replicated comparison at a bed-constrained configuration, mean inpatient admission waiting time is 14.8 days in the hybrid model and 25.8 days in an otherwise identical DES-only model (42% lower, $p < 0.001$), and 25%–64% lower under two further conditions.

[3] **M. Baghi**. "Conditional Validity for Adaptive Modality Acquisition: When the Policy Chooses Its Own Calibration Group." *Manuscript under review.* [arXiv:2608.15520] [code]

- Certifies a prediction when the modalities already observed determine the decision, and otherwise abstains and recommends which modality to acquire next. Provides finite-sample guarantees conditional on the terminal modality pattern created by adaptive acquisition, giving coverage conditional on the terminal pattern rather than marginal coverage across patterns. Evaluated across four multimodal datasets spanning ECG, speech and wearable sensing (PTB-XL, CMU-MOSEI, MHEALTH, IEMOCAP), with reproducible experiments and public code.

[4] **M. Baghi**, X. Liu, K. Paynabar. "A Product-Grassmann Tucker Surrogate for Parametric Field Prediction." *Submitted to the Journal of Computational Physics.* [arXiv:2610.34559]

- Develops a Tucker-based surrogate for parameterized spatiotemporal simulation fields that keeps the spatial directions and time as separate modes: mode-wise factor subspaces are interpolated on a product Grassmann manifold, orthogonal Procrustes alignment places the Tucker cores in one coordinate system, and a truncated SVD models their variation. Across five PDE benchmarks in two and three spatial dimensions, including public CFD data, reduces the interpolation size by 13×–482× and the stored model by 13×–464× relative to a single-subspace Grassmannian surrogate, with 3.3×–8.5× faster cached queries and mean errors within 0.22 percentage points; in a 3D grid study at 64^3 the interpolation-size ratio reaches 2,260× and the query speedup 11.9×.

[5] **M. Baghi**, X. Liu, K. Paynabar. "Supervised Active-Subspace Input Reduction and Risk-Controlled Field Uncertainty for Tensor-Valued Computer Experiments." *Completed manuscript, ready for submission.*

- Reduces tensor-valued simulation inputs through supervised active subspaces, represents the output via a higher-order decomposition, and places a decomposition-based Gaussian process on the reduced map to deliver field-level predictive uncertainty, with finite-sample subspace-estimation guarantees via matrix perturbation theory (Wedin / Davis–Kahan).

[6] **M. Baghi**. "Pooling and Drift in Delayed Bandits." *Manuscript under review.* [arXiv:2609.01761] [code]

- Addresses bandit feedback in which an action's loss arrives only after a fixed delay. State-EXP3 charges one delayed outcome to every action that could have produced the observed state; the estimation cost is an effective dimension set by the overlap between rows of the action-to-state matrix, not by the number of actions or states. Adds the delayed and noisy setting, the coarsening trade-off, and a drift lower bound showing the cost of delayed feedback cannot be removed by pooling. Reduces regret by up to 79% against action-level weighting and by 32–68% against a rate-optimal action-only delayed-bandit baseline in synthetic experiments.

Working Papers and In Preparation

[7] **M. Baghi**, K. Paynabar, X. Liu. "Reliability-Calibrated Stability for Continual Multimodal Representation Learning." *Research proposal; manuscript in preparation.*

- Argues that uniform stability in continual multimodal learning is misspecified when historical cross-modal alignment mixes semantic shared structure with spurious correlations. Proposes a perturbation-based spectral-block reliability estimator on the Grassmannian and a reliability-weighted projection operator for the alignment head, with an excess-risk decomposition identifying the cost of preserving nuisance blocks vs. releasing shared blocks.

[8] "Reference-Measure Conformal Prediction for Subspace-Valued Outputs." *Manuscript in preparation.*

- Distribution-free conformal prediction sets for outputs that are subspaces rather than vectors, built against a reference measure on the Grassmannian.

[9] "Topological Conformal Risk Control: Distribution-Free Uncertainty for Persistent Features." *Manuscript in preparation.*

- Conformal risk control for persistent homology features, giving distribution-free guarantees on topological summaries.

Research Experience

Georgia Institute of Technology *Atlanta, GA*
Graduate Research Assistant — Prof. Kamran Paynabar & Prof. Xiao Liu *Aug 2023 – Present*

- **Geometry-Aware Reduced-Order Modeling for Parametric PDE Systems.** Developed novel statistical learning frameworks for constructing fast surrogates of computationally expensive high-fidelity simulations parameterized by PDEs. Proposed stable injective mappings from the Grassmann manifold to a constrained Euclidean space, enabling classical regression and uncertainty quantification over spaces of orthogonal subspaces. Integrated gradient boosting trees (cXGBoost) and projected Gaussian processes as hybrid predictors; on four parametric problems (cylinder wake, Kolmogorov flow, the Kuramoto–Sivashinsky equation and a forced beam), cXGBoost has lower mean reconstruction errors than the projected Gaussian process and Grassmann interpolation. Open-source implementation: `github.com/melikabaghi/cXGBoost`.
- **Tensor-Structured Surrogate Modeling on Product Grassmann Manifolds.** Developed a surrogate framework preserving the native multiway structure of spatiotemporal simulation fields. Combines Tucker decomposition of each training snapshot, product-Grassmann interpolation of mode-wise factor subspaces, orthogonal Procrustes core alignment, and a Core-of-Cores low-rank model for aligned-core variation. Across five PDE benchmarks, reduces the interpolation size 13×–482× and speeds up cached queries 3.3×–8.5× relative to a single-subspace Grassmannian surrogate, with mean errors within 0.22 percentage points (arXiv:2610.34559).
- **Reliability-Calibrated Continual Multimodal Representation Learning.** Identified that uniform stability in continual multimodal learning can entrench spurious cross-modal shortcuts. Proposed reliability-calibrated stability: stores a compact spectral-block memory of cross-modal covariance, estimates block reliability via perturbation stability on the Grassmannian, refines reliability when a new modality arrives using cross-covariance evidence, and applies reliability-weighted gradient projection to the alignment head. Derived an excess-risk decomposition under a linear-Gaussian model with block subspace separation, perturbation informativeness, and eigengap assumptions.
- **Directed Projects (Georgia Tech, 2023–2025).** In the course of doctoral study, completed three supervised applied projects under Profs. Paynabar and Liu: (i) *Sensitivity-Guided Diffusion Models*: designed diffusion-based generative models constrained to physically meaningful latent subspaces via sensitivity-informed perturbation schedules, implemented in PyTorch; (ii) *Multimodal Phenotype Modeling*: built representation learning pipelines integrating single-cell transcriptomics, longitudinal EHR data, and imaging-derived phenotypes using contrastive learning and transformer-based phenotype extraction from clinical notes; (iii) *Drug Safety and Clinical Risk Modeling*: developed a temporally-aware adverse drug event detection framework from longitudinal EHR data using regularized regression, tree ensembles, and neural embeddings, with calibration and fairness analyses across demographic strata.

Amirkabir University of Technology *Tehran, Iran*
Graduate Research Assistant — Prof. Hadi Mosadegh *Aug 2021 – Jul 2023*

- **Simulation-Based Scheduling Optimization via Hybrid ABS+DES and Machine Learning.** Modeled patient treatment journeys via a hybrid discrete event simulation (DES) and agent-based simulation (ABS) framework in AnyLogic. Represented registration, consultation, admission, bed allocation, and discharge through DES; represented patient, physician, and ward behaviors through agent-based logic including medication adherence, dropout, doctor switching, and emergency escalation. Conducted a 256-run two-level fractional factorial design over 16 operational factors, which identified bed capacity as the most consistent driver of waiting. Against an otherwise identical DES-only model, the hybrid model's mean inpatient admission waiting time was 42% lower (14.8 vs. 25.8 days, $p < 0.001$).

Amirkabir University of Technology *Tehran, Iran*
Undergraduate Research Assistant — Prof. Mehdi Sabzeh Parvar *May 2018 – May 2020*

- **Learning-Based Safety Region Estimation for Nonlinear Dynamical Systems.** Developed a six-

degree-of-freedom physics-based simulator in MATLAB/Simulink using system velocity equations and operational constraints, and generated training and test datasets across multiple operating scenarios. Implemented machine learning models in Python, including a three-layer neural network, to predict system behavior and expand the feasible operating range under hybrid inputs; studied nonlinear behavior and distributional robustness under input uncertainty.

Teaching Experience

Georgia Institute of Technology *Atlanta, GA*
Teaching Assistant, then Head Teaching Assistant — ISyE 6525 (Prof. Paynabar) *Aug 2023 – Present*

- Lead and coordinate a TA team for a graduate course of 150+ students per semester covering nonparametric methods, dimensionality reduction, sparse modeling, manifold learning, regularized regression (OLS, Ridge, LASSO, Adaptive LASSO, Elastic Net), and high-dimensional inference.
- Design advanced problem sets and final projects emphasizing theoretical understanding, reproducible ML workflows, and model validation under distribution shift.
- Mentor student research projects on uncertainty quantification and interpretability; conduct weekly office hours and lead review sessions.
- Recognized as **Outstanding Online TA** (2025, Tech to Teaching Program, Georgia Institute of Technology).

Amirkabir University of Technology, Industrial Engineering *Tehran, Iran*
Teaching Assistant — Principles of Simulation (Prof. Abbas Ahmadi) *Sep 2022 – Aug 2023*

- Taught class sessions explaining Python and simulation code in MATLAB, Simulink, Arena, and AnyLogic for Monte Carlo methods, discrete event simulation, and stochastic process modeling. Held office hours and evaluated student projects, including healthcare operations simulations.

Amirkabir University of Technology, Aerospace Engineering *Tehran, Iran*
Teaching Assistant — Engineering Dynamics (Prof. Hamid Reza Ovesy) *Feb 2020 – Aug 2023*

- Delivered scheduled lectures and recitations on analytical mechanics, Lagrangian dynamics, and stability analysis. Designed and administered exams.
- **Ranked #1 Teaching Assistant in the department** by student evaluation for three consecutive years.

Professional Experience

Royan Institute *Tehran, Iran*
Course Instructor — Artificial Intelligence Laboratory *Aug 2022 – Feb 2023*

- Designed and delivered a project-based AI curriculum covering rule-based expert systems, hybrid architectures, and applied machine learning in clinical healthcare contexts. Supervised end-to-end student projects from problem formulation through evaluation.

Arvand Pishro (Automotive R&D) *Safadasht, Iran*
Research Intern — R&D Division *Sep 2019 – Dec 2020*

- Designed and validated a closed-loop fuel tank cooling and monitoring system integrating sensor feedback and automated temperature control. Collaborated with multidisciplinary engineering teams to translate requirements into validated prototype systems.

Honors and Awards

- **Finalist**, INFORMS Case Competition, for the teaching case "One Queue, Two Worlds: Managing Hospital Capacity for Medical Tourists and Local Patients" (sole author; presenting at the INFORMS Annual Meeting,

November 2026) 2026

- **Most Innovative Approach Award**, NSF Future Manufacturing Data Challenge 2026
- **Conference Fund Award**, Student Government Association, Georgia Institute of Technology — with College of Engineering matching funds 2026
- **Third-year Progress Travel Award**, H. Milton Stewart School of Industrial & Systems Engineering, Georgia Institute of Technology 2026
- **Travel Award**, Institute for Mathematical and Statistical Innovation (IMSI) — an NSF-funded institute 2025
- **Outstanding Online TA Award**, ISyE 6525, Tech to Teaching Program, Georgia Institute of Technology 2025
- **Conference Fund Award**, Student Government Association, Georgia Institute of Technology 2024
- **Stewart Topper Fellowship**, Georgia Institute of Technology 2023
- **First-year Fellows Travel Award**, H. Milton Stewart School of Industrial & Systems Engineering, Georgia Institute of Technology 2023
- **Mansour Doctoral Fellowship**, North Carolina State University — granted for exceptional academic promise in Industrial Engineering 2023
- Direct entry to M.S. in Industrial and Systems Engineering as an elite student, ranked **3rd of 120** applicants, Amirkabir University of Technology 2021
- Direct entry to M.S. in Aerospace Engineering as an elite student, ranked **2nd of 120** applicants, Amirkabir University of Technology 2021
- Selected as an elite student and approved to pursue a dual degree in Aerospace Engineering, Amirkabir University of Technology 2018
- Ranked in the **top 1%** among 150,000+ participants in the National University Entrance Exam (Physics & Mathematics track) 2016

Technical Skills

- **Programming & Data:** Python, SQL, R, PySpark, NumPy, pandas, SciPy, MATLAB
- **Machine Learning:** PyTorch, scikit-learn, XGBoost, deep learning, diffusion generative models, gradient boosting, Gaussian processes, contrastive and representation learning, multimodal learning, multi-armed bandits and online learning, LLM-based information extraction, ensemble methods
- **Statistics & Modeling:** Statistical modeling, nonparametric statistics, design of experiments, uncertainty quantification, calibration, conformal prediction, high-dimensional inference, statistical learning theory, stochastic processes, probabilistic modeling
- **Mathematics & Scientific Computing:** Optimization on manifolds, Riemannian geometry (Grassmann, Stiefel, product manifolds), tensor decompositions (Tucker, HOSVD), reduced-order and surrogate modeling, computational fluid dynamics (CFD), finite element analysis (FEA), Lattice Boltzmann methods, Monte Carlo simulation
- **Optimization & Simulation:** Mixed-integer programming, AnyLogic (ABS + DES), Arena, Simulink, Ansys Fluent, GAMS
- **Tools & Computing:** Git, Docker, Linux, AWS, Azure, cloud computing, GPU/HPC, SLURM, LaTeX

Selected Academic Projects

- **LLM Agents as Simulated Patients in a Queueing Study:** Prompted three dated OpenAI model snapshots through the API, with a fixed prompt and a strict JSON schema, to give choice probabilities for 36 synthetic patient profiles; cached every response and calibrated each model to the same baseline rates. Used

each model as the patient behavior in a queue simulation of eight capacity policies (120 paired seeds): the models agreed at baseline but not on the best policy under a demand surge (relative regret 37% to 52%). A follow-up with 2,160 sampled LLM actions found that cheap behavioral rules picked the same policy, so the study stopped at its go/no-go gate. *Python, OpenAI API* (2026)

- **Interaction Tests and Survival Models on TCGA Multi-Omics Data:** Built kernel permutation tests for interaction among four omics layers (RNA-seq, miRNA, copy number, methylation); in Monte Carlo simulations the test held its nominal 5% type-I rate, and all ten TCGA cancers rejected the null ($p < 0.005$), also after batch correction by submitting site. Added a patient-level score to Cox models with clinical covariates (glioma HR 1.80 per SD, 95% CI 1.51 to 2.13); its C-index gain did not hold on held-out sites. *Python, lifelines* (2026)
- **Single-Cell and Spatial Transcriptomics Pipeline for Breast Cancer:** Built a nine-stage pipeline on a simulated five-patient scRNA-seq and Visium cohort (QC, integration, clustering, marker-based annotation with 88.3% accuracy, cell2location deconvolution); scored ligand-receptor signaling per spatial niche against a permutation null with FDR control (6 of 8 predicted target programs up, $p = 0.001$); reran the pipeline on a public 10x Visium tumor slide. *Python, Scanpy, scvi-tools, cell2location, Squidpy, PyTorch* (2026)
- **Random-Effects Meta-Analysis with a Coverage Simulation:** Pooled 13 clinical BCG vaccine trials (RR 0.49, 95% CI 0.33 to 0.73, $I^2 = 92\%$); a latitude meta-regression explained 76% of the between-trial variance. In 18,000 simulated meta-analyses with 5 trials, Hartung-Knapp intervals covered the true effect 92% of the time against 84% for Wald intervals. *R, metafor* (2026)
- **Probing Language-Model Representations:** Trained probes at three depths on residual-stream activations of Pythia language models (70M to 1.4B parameters) and tested every finding against shuffled-label nulls and a matched control family; day-of-week order emerged from 160M parameters up (rank correlation about 0.91) and never in controls. *Python, PyTorch, Transformers* (2026)
- **Conformal Prediction on Graphs:** Benchmarked learned graph-diffusion corrections for conformal prediction sets (scalar, diagonal and full sheaf edge operators) on a two-layer GCN across six heterophilic graph datasets against APS and DAPS baselines; all methods held 90% coverage, but the sheaf operators gave no set-size gain beyond seed noise, so a stopping rule set in advance ended the idea. *Python, PyTorch* (2026)
- **Unit Commitment and Economic Dispatch (course project):** Formulated a mixed-integer unit commitment model that chooses which generating units run and at what output, minimizing total cost within each unit's capacity limits. *GAMS*
- **Predictive Maintenance and Anomaly Detection in Industrial Systems:** Developed an end-to-end machine learning pipeline for sensor-based fault detection, integrating time-series forecasting, feature engineering, and real-time monitoring. Applied ARIMA and LSTM models for trend prediction and ensemble classifiers for anomaly detection in multivariate telemetry; achieved over 15-hour advance fault warnings through hybrid Weibull–state-space degradation modeling with online retraining. Deployed the workflow using Spark for scalable, low-latency inference across industrial sensor networks. *Python, PySpark* (2025)
- **Sensitivity-Guided Diffusion Models for Scientific Dynamical Systems:** Designed diffusion-based generative models constrained to physically meaningful latent subspaces derived from spectral decomposition and sensitivity analysis. Derived sensitivity-informed directional perturbation schedules guiding the forward SDE along high-variance manifold directions to improve sample efficiency and structural fidelity in data-scarce scientific regimes. *Python, PyTorch* (2024–2025)
- **Multimodal Phenotype Modeling for Translational Disease Discovery:** Constructed representation learning pipelines integrating single-cell transcriptomics, longitudinal EHR data, and imaging-derived phenotypes for disease-relevant biological variation modeling. Built structured cellular and patient embeddings using contrastive representation learning and dimensionality reduction; leveraged transformer-based language models for structured phenotype extraction from unstructured clinical notes. *Python, PyTorch* (2024–2025)
- **Machine Learning for Drug Safety and Clinical Risk Modeling:** Developed an adverse drug event detection framework using temporally-aware patient embeddings integrating medication trajectories, laboratory values, and comorbidity profiles from longitudinal EHR data. Applied regularized regression, tree-based ensembles, and neural representation-learning models; conducted calibration, fairness, and robustness analyses across demographic strata. *Python* (2024–2025)
- **Tensor-Structured Learning Pipelines for High-Dimensional Scientific Data:** Designed deep and tensor-structured learning architectures for parameterized scientific fields; investigated representation learning under structural and geometric constraints. *Python, PyTorch* (2024)

- **Internet Access and Female Economic Empowerment: A Cross-National Nonparametric Analysis:** Studied the relation between national internet usage and two indicators of female economic empowerment (employment rate and share of senior and middle management positions) across 92 countries using World Bank Gender Data Portal indicators for 2022. Classified nations by internet access ($\geq$50% vs. $<$50% of population), assessed normality via Shapiro–Wilk and Anderson–Darling tests with histogram and Q–Q diagnostics, and applied two Mann–Whitney U tests with a Bonferroni correction.
- **Anomaly Detection in Sensor Streams:** Built ARMA-based statistical monitoring with sliding-window control charts to flag faults in industrial sensor streams early, trading off detection delay against false-alarm rate. *Python* (2023)
- **Loan Default Prediction:** Trained and compared ensemble classifiers for consumer credit risk, evaluating ranking performance alongside probability calibration rather than accuracy alone. *Python* (2023)
- **Food Recommender Expert System:** Designed and implemented a knowledge-based recommendation system tailored to individual health characteristics and dietary needs. *Python* (2021–2022)
- **Breast Cancer Prediction:** Built and statistically evaluated diagnostic classification pipelines, comparing generalization across models rather than training accuracy. *Python, scikit-learn* (2021–2022)
- **Airfoil Flow Simulation and CFD Analysis:** Meshed a supersonic airfoil in Gambit and simulated the flow in Ansys Fluent to capture the shock. *Gambit, Ansys Fluent* (2021–2022)
- **Safe Operating Range under Hybrid Inputs:** Built a six-degree-of-freedom physics-based simulator in MATLAB/Simulink from system velocity equations and operational constraints, generated data across operating scenarios, and trained neural network models in Python to predict system behavior and expand the feasible operating range. *MATLAB, Simulink, Python* (2018–2020)
- **Nurse Scheduling with Quick-Response Methods:** Formulated and solved mixed-integer programming models for healthcare workforce scheduling under demand uncertainty. *GAMS* (2020–2021)
- **Healthcare Operations Quality Improvement:** Applied Six Sigma methodology to identify and reduce service quality deficiencies in a healthcare organization. (2021–2022)

Service and Outreach

- Core team member, CANSAT satellite design competition, Amirkabir University of Technology 2019
- Staff organizer, National AerUp Competition, Amirkabir University of Technology 2017, 2018
- Referee, National Aerospace Competitions, Aerospace Engineering Department, Amirkabir University of Technology 2017, 2018

Languages

- **Persian:** Native
- **English:** Excellent (TOEFL iBT 110 / 120: Reading 28, Listening 30, Speaking 25, Writing 27)
- **Korean:** Excellent spoken proficiency